PROJECT REPORT

AI Powered GSR/EDA Counselling Bio-Feedback System

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Note: MannPulseAI is a research / counselling-support prototype and is not a diagnostic medical device.

1. Introduction

Mental health and emotional well-being have become increasingly critical concerns in India, where stress, anxiety, and related psychological challenges affect a growing proportion of the population across age groups and professions. Despite this rising need, access to objective, data-driven support tools within the counselling and neuropsychiatric ecosystem remains extremely limited. Counselling practice continues to rely almost entirely on subjective self-report and verbal disclosure from clients — counsellors depend on what a client chooses to share and how well they can articulate their internal state, yet physiological stress responses often occur beneath conscious awareness, meaning clients may not always recognize or verbalize the true intensity of their own arousal or distress.

At the same time, affordable, objective physiological feedback tools that could bridge this gap remain largely unavailable to small clinics, independent counsellors, and educational institutions in India. Existing physiological monitoring and biofeedback systems tend to be expensive and designed for high-end clinical or research settings, placing them out of reach for the vast majority of practitioners who serve everyday clients.

MannPulseAI — guided by the philosophy "Measure. Understand. Heal." — is proposed as a direct response to this problem. It is an AI-powered Galvanic Skin Response (GSR) / Electrodermal Activity (EDA) counselling biofeedback system developed by Team Debuggers Squad as a low-cost, portable alternative to conventional physiological monitoring systems. A simple finger-worn sensor captures electrodermal activity — a well-established marker of sympathetic nervous system arousal — during a counselling session. This raw signal is processed through baseline calibration, feature extraction, and machine learning-based analysis to produce a real-time, interpretable measure of the client's stress and arousal state.

Built using accessible hardware — an Arduino/ESP32 microcontroller, a finger GSR sensor, and a laptop-based dashboard — MannPulseAI has an estimated cost of approximately ₹2,700 per unit, making it far more accessible than premium physiological monitoring systems and suitable for small clinics, college wellness centers, and academic research labs across India.

The dashboard visualizes the live arousal graph, session timer, and a Stress-Arousal Index, alongside a machine learning-derived Total Recovery/Response (TRR) score computed using Random Forest, Ridge Regression, and Exponentially Weighted Moving Average (EWMA) models. This gives counsellors insight into both the client's current arousal level and their recovery trend across a session. At the end of each session, the system automatically generates a concise PDF summary report for quick reference and inclusion in session notes.

MannPulseAI is explicitly positioned as a **research and counselling-support prototype, not a diagnostic medical device** — intended to augment, not replace, professional clinical judgment. This distinction is maintained throughout its design, testing, and roadmap, which includes pursuing clinical validation and regulatory pathways before any diagnostic claims are made.

- ## Motivation Behind the Project

The idea for MannPulseAI emerged from a simple but important observation: while wearable health technology has advanced rapidly for fitness and cardiac monitoring, very little of that innovation has been directed toward affordable, session-level tools for mental health practitioners in India. Global biofeedback systems used in premium clinics can cost anywhere from tens of thousands to lakhs of rupees, and are rarely designed with the workflow of a small counselling practice in mind. Team Debuggers Squad set out to explore whether a stripped-down, purpose-built version of this technology — focused specifically on session-wise GSR/EDA monitoring rather than broad clinical diagnostics — could be built at a fraction of the cost while still delivering genuinely useful, real-time insight to a counsellor.

This motivation shaped every subsequent design decision in the project: prioritizing low-cost, widely available hardware components over specialized medical-grade sensors; favoring simplicity and portability over an exhaustive feature set; and building a machine learning pipeline that produces clear, interpretable outputs rather than opaque diagnostic scores. The result is a prototype that is deliberately scoped to be a support tool rather than a replacement for clinical expertise.

2. Problem Statement

- ### Affordable Stress-Arousal Feedback Gap in India

Counselling and neuropsychiatry still rely heavily on subjective self-report, while affordable, objective physiological feedback tools remain largely unavailable in India. This creates several compounding problems that limit the quality and consistency of counselling support:

#	Core Problem Area	Description
1	Subjective dependency	Many counselling sessions depend entirely on self-report and verbal disclosure from the client.
2	Lack of affordable tools	Small clinics and independent practitioners lack access to affordable physiological monitoring tools.
3	Missed signals	Manual observation by counsellors often misses hidden or subtle stress/arousal responses.
4	Inconsistent tracking	Progress tracking across multiple sessions is limited and inconsistent.
5	Restricted insight	Limited access to physiological data restricts data-driven counselling and outcome insights.

- ### Four Root Causes Identified

a) **Cost Barrier:** Conventional physiological monitoring systems and wearables are expensive, putting them out of reach for most Indian clinics and patients.

b) **Subjective Observation:** Many sessions rely purely on verbal disclosure and self-report, with no objective corroboration.

c) **Small Clinic Gap:** Counsellors in smaller practices need simple, live biofeedback tools to support their clinical insight.

d) **Progress Tracking Need:** There is a clear need for session-wise stress and recovery tracking over time.

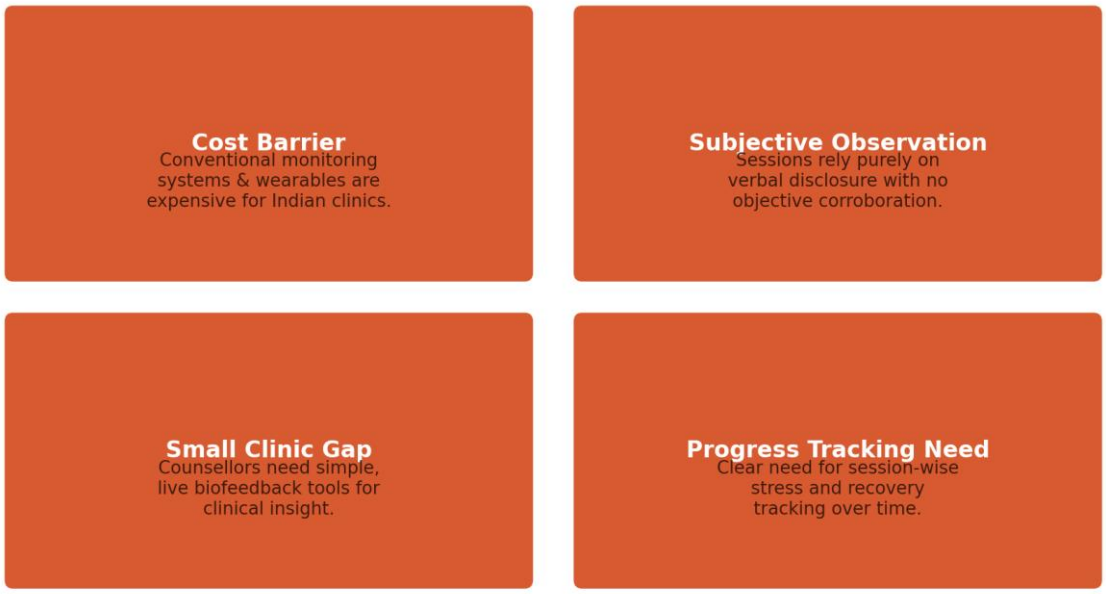


Fig. 1 — The four root causes behind the affordable stress-arousal feedback gap

- **Identified Need:** A portable, low-cost, AI-assisted GSR biofeedback system that visualizes stress arousal in real time, compares data across sessions, and generates quick reports to support the counselling process. MannPulseAI positions itself directly against this accessibility gap by offering an affordable, portable, session-wise biofeedback workflow.

- ### Why This Problem Matters Now

India's counselling and mental health infrastructure is expanding rapidly, with a growing number of college wellness centers, private practices, and community mental health initiatives. However, this expansion has largely occurred without a corresponding increase in access to objective, physiological assessment tools. As demand grows, counsellors are increasingly expected to demonstrate measurable outcomes and track client progress systematically — something that is very difficult to do consistently using self-report alone. MannPulseAI addresses this timing gap by offering a practical, low-cost tool that can be adopted immediately within existing counselling workflows, without requiring specialized clinical infrastructure or extensive retraining.

3. Technical Approach & Methodology

The system methodology follows a seven-stage pipeline from raw signal collection to auto-generated clinical reporting:

1. **GSR Data Acquisition:** A finger-based GSR/EDA sensor captures sweat-gland activity during the counselling session.
2. **Signal Conditioning & Baseline Calibration:** The raw signal is cleaned, smoothed, and baseline-calibrated for individual physiological stability.
3. **Feature Extraction:** 16 physiological features are derived, including average arousal, peak arousal, recovery trend, and signal quality.
4. **Microcontroller-Based Processing:** An Arduino/ESP32 microcontroller streams processed data to a laptop in real time.
5. **Real-Time Visualization:** A live dashboard displays the arousal graph, current status, and a session timer.

6. **ML-Based Analysis:** Random Forest, Ridge Regression, and Exponentially Weighted Moving Average (EWMA) models are used to estimate a Total Recovery/Response (TRR) score and trend.
7. **Auto Clinical Report Generation:** A quick PDF session summary is produced automatically with key insights.

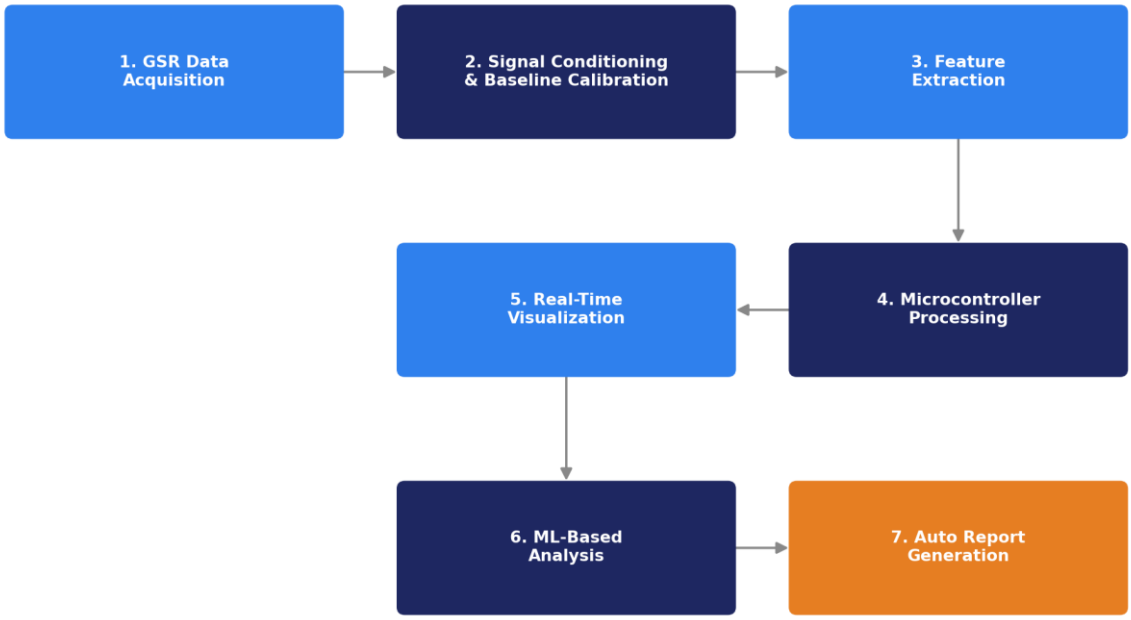


Fig. 2 — Seven-stage technical processing pipeline

• Hardware Components

- Finger GSR sensor (worn on index and middle fingers)
- Portable hardware prototype (custom enclosure housing the Arduino/ESP32 and sensor interface)
- Laptop dashboard (Python/PyQt-based software interface)

• Signal Processing Pipeline in Detail

The signal processing stage is central to the reliability of MannPulseAI's output. Raw electrodermal signals are inherently noisy, influenced by factors such as ambient temperature, sensor contact pressure, and individual differences in baseline skin conductance. To address this, the system applies a multi-step conditioning process: first, a moving-average filter smooths high-frequency noise; second, a short calibration period at the start of each session establishes an individual baseline against which subsequent readings are normalized; and third, outlier rejection removes transient artifacts caused by movement or momentary sensor displacement.

Once conditioned, the signal is decomposed into 16 physiological features spanning both time-domain statistics (mean, peak, variance of arousal) and trend-based measures (rate of recovery after a peak, frequency of arousal spikes, and overall session stability). These features form the input to the machine learning layer, ensuring that the downstream TRR score reflects meaningful physiological patterns rather than raw sensor noise.

• Software Stack

The software layer is built primarily in Python, chosen for its strong ecosystem of scientific computing and machine learning libraries. The dashboard interface uses PyQt to provide a responsive, native desktop experience capable of rendering live graphs at interactive frame rates. The machine learning models are implemented using scikit-learn, and the automated report generator uses a lightweight PDF-writing library to assemble session summaries without requiring manual formatting.

- **Sample ML Analysis Output**

A representative session output includes a TRR Score of 72 out of 100 (rated "Good"), an improving trend indicator, and model attribution showing that the score was derived from the combined output of the Random Forest, Ridge Regression, and EWMA models. This multi-model approach was chosen deliberately: Random Forest captures non-linear relationships between features, Ridge Regression provides a stable, interpretable baseline estimate, and EWMA smooths the trend signal across the session to reduce the impact of short-term fluctuations.

- **Sample Session Summary Report Fields**

- Patient ID, Session Date/Time
- Metrics: Average Arousal, Peak Arousal, Recovery Trend, TRR Score, Session Quality
- Overall Insight (auto-generated natural-language summary)

Key Outputs: real-time stress monitoring, ML-based TRR scoring, session-wise tracking, 1-minute auto-generated PDF reports, a patient ID system, and structured counselling support data.

4. System Architecture and Workflow

The end-to-end data flow moves sequentially from physical sensor input through processing and analysis to a final report artifact, as shown below:

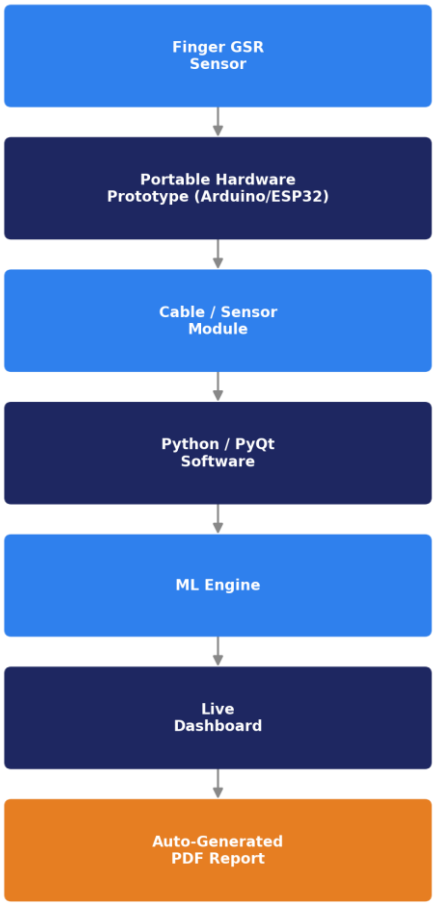


Fig. 3 — End-to-end system data flow

➢ Software Dashboard Features

- Live GSR/Arousal stream graph (µS over time)
- Session timer and live/paused status indicator
- Stress-Arousal Index (0–100 scale) with qualitative label (e.g., "Calm," "Moderate")
- TRR Score gauge with trend arrow (Improving/Declining/Stable)
- HRV (Heart Rate Variability) Index as a supporting balance indicator
- Navigation panel: Overview, Live Graph, Analytics, Sessions, Reports, Settings
- Action controls: Live, Compare, Notes, Export

➢ Low Hardware Complexity

The hardware chain intentionally uses commonly available components, keeping the system easy to assemble, demonstrate, and maintain — an important design decision for scalability into resource-limited clinics.

➢ Data Handling and Privacy Considerations

Because the system handles physiological data tied to individual clients, data handling practices have been considered from an early stage even at the prototype level. Session data is stored locally on the counsellor's laptop rather than transmitted to any external server, minimizing exposure of sensitive information. Each session is associated with a Patient ID rather than personally identifying details within the raw dataset, and generated PDF reports are intended to be handled under the same confidentiality standards a counsellor would already apply to session notes. As the system matures toward pilot deployment, the team intends to formalize these practices against recognized data protection frameworks.

5. Prototype Testing and Validation

➢ Live Testing

The team conducted real-world demo sessions, including demo testing with volunteers, real finger-sensor signal capture, and live signal visualization on the dashboard. Testing was carried out at events, colleges, and counselling engagements, using volunteer-based signal testing sessions involving students, professionals, and mental health volunteers. Real sensor placement and capture protocols were followed using the MannPulseAI finger GSR sensor to ensure accurate physiological signal collection.

➢ Public Engagement and Social Proof

Demo content shared publicly reached over 2.4K+ views, with approximately 1K+ likes and 2K+ shares. Public feedback included positive responses from students, professionals, and mental health advocates; high interest in affordable stress monitoring; and requests for early access and demonstrations across multiple cities.

➢ Testing Highlights Summary

Highlight	Description
Live prototype demonstrations	Conducted at events, colleges, and counselling engagements
Volunteer-based signal testing	Real sessions with students, professionals, and mental health volunteers
Real sensor placement and capture	Accurate physiological signal capture using the finger GSR sensor
Public showcase and feedback	Demos and prototypes shared across platforms for community insight

Highlight	Description
Early user interest and visibility	Strong traction, engagement, and validation from target communities

➢ Testing Environment and Conditions

Testing sessions were conducted in a range of ambient conditions to assess signal stability outside a controlled lab environment — including indoor demonstration booths, classroom settings, and informal one-on-one setups. This variability was intentional: since MannPulseAI is designed for everyday counselling clinics rather than climate-controlled laboratories, it was important to confirm that the finger sensor maintained usable signal quality across normal indoor temperature and humidity ranges, and that the dashboard's baseline calibration step adequately compensated for these differences between sessions.

Validation Status Achieved: demo-ready prototype, real signals collected, active public engagement, and early validation signals — with documentation retained for presentation evidence and prototype validation context.

6. Impact and Benefits

MannPulseAI delivers eight core benefit areas as a low-cost, AI-powered GSR/EDA counselling biofeedback system:

a) **Objective Counselling Support:** Helps counsellors observe physiological arousal during sessions rather than relying solely on self-report.

b) **Affordable Indian Prototype:** Low-cost hardware paired with high-value software functionality.

c) **Useful for Stress & Anxiety Monitoring Support:** Provides non-diagnostic, session-level feedback.

d) **Real-Time GSR Visualization:** Displays live stress-arousal graphs during the session.

e) **Session-Wise Tracking:** Enables comparison of arousal and recovery patterns across repeated sessions.

f) **Portable and Easy to Demonstrate:** Compact setup suitable for clinics, labs, and live demos.

g) **Education and Research Value:** Useful for learning, experimentation, and validation studies.

h) **Future Clinical Pathway:** Can progress toward broader, clinically validated use following formal approvals.

i) **Key Outcomes:** physiological insight generation, biofeedback support for clients, real-time monitoring, session-to-session comparison, portability for field demonstration, and 1-minute automated reporting.

➢ Who Benefits and How

Stakeholder	Benefit
Counsellors	Gain an objective data point to complement clinical judgment and track client progress consistently
Clients	Receive session feedback that can improve self-awareness of stress patterns
Clinics	Can offer a differentiated, technology-supported counselling service at low added cost
Students / Researchers	Gain hands-on access to affordable physiological data for learning and experimentation

7. Feasibility and Viability

➢ Cost and Positioning

- **Estimated Hardware Cost:** ₹2,700 approx.
- **Positioning:** Non-diagnostic counselling-support prototype designed for clinics, counsellors, students, and research setups.

➢ Indicative Hardware Cost Breakdown

Component	Approx. Cost (₹)
Arduino / ESP32 microcontroller	700
Finger GSR/EDA sensor	600
Cable / sensor interface module	300
Enclosure and connectors	500
Miscellaneous (wiring, assembly)	600
Estimated Total	≈ 2,700

➢ Three Pillars of Feasibility

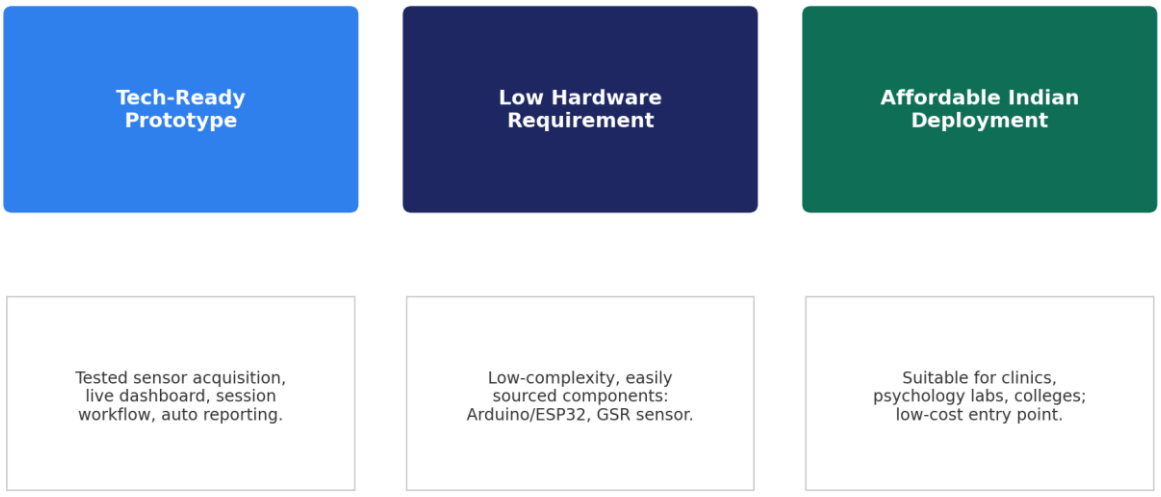


Fig. 4 — Three pillars supporting the feasibility of MannPulseAI

- **Tech-Ready Prototype:** Tested finger-sensor signal acquisition, a working live laptop dashboard, a functioning session-wise data workflow, and automatic report generation.
- **Low Hardware Requirement:** The build uses low-complexity components (Arduino/ESP32, finger GSR sensor, cable/sensor module, Python/PyQt software, ML engine) that are easy to assemble, demonstrate, and source.
- **Affordable Indian Deployment:** Suitable for counselling clinics, psychology labs, and colleges; offers a lower-cost entry point for data-driven counselling support; and is useful for demonstrations, education, and early institutional adoption.

➢ Practical Value Points

Portable kit design, real-time stress graph, TRR score feedback, session-wise tracking, and easy installation.

➢ Scalability Potential Across Use-Case Segments

Segment	Value Delivered
A. Counselling Clinics	Enhances session quality with objective stress insights
B. Neuropsychiatry Support	Supports treatment monitoring and therapeutic feedback
C. Medical Colleges	Useful for training, curriculum, and clinical exposure
D. Research Labs	Enables low-cost physiological research and data collection
E. Student Innovation	Encourages student projects, startups, and proof-of-concept solutions
F. Future Pathway	Scalable to cloud platforms, mobile apps, and AI-driven analytics

8. Market Scalability and Business Model

Primary Target Customers: Neuropsychiatrists, psychologists and counsellors, mental wellness centers, medical colleges, and research labs.

➢ Scalability Path

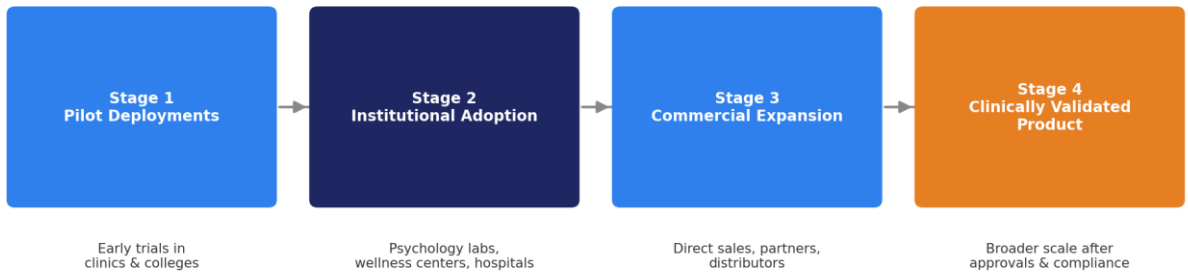


Fig. 5 — Four-stage scalability path from pilot to validated product

Stage	Focus
Stage 1	Pilot Deployments — early trials in clinics and colleges
Stage 2	Institutional Adoption — psychology labs, wellness centers, hospitals
Stage 3	Commercial Expansion — direct sales, partners, distributors
Stage 4	Clinically Validated Product — broader scale following approvals and compliance

➢ Business Model Components

Device sales, software/dashboard access subscriptions, auto-report and analytics value-add services, training and installation services, annual service/maintenance contracts, and institutional partnerships.

➢ Go-to-Market Channels

- Official website
- Direct outreach to clinics and hospitals
- Distributor / dealer network
- Medical demonstrations and conferences
- Online marketplace (later-stage channel)

➢ Why It Can Scale

- Low-cost and portable — accessible across India
- Easy laptop/software integration — no complex infrastructure needed
- Useful for counselling, education, and research — wide application scope
- Made in India — scalable globally
- Strong future potential following clinical validation and regulatory approval

Growth Vision: Prototype → Pilot → Institutional Rollout → Validated Product.

9. Roadmap and Future Work

➢ Seven-Stage Development Roadmap

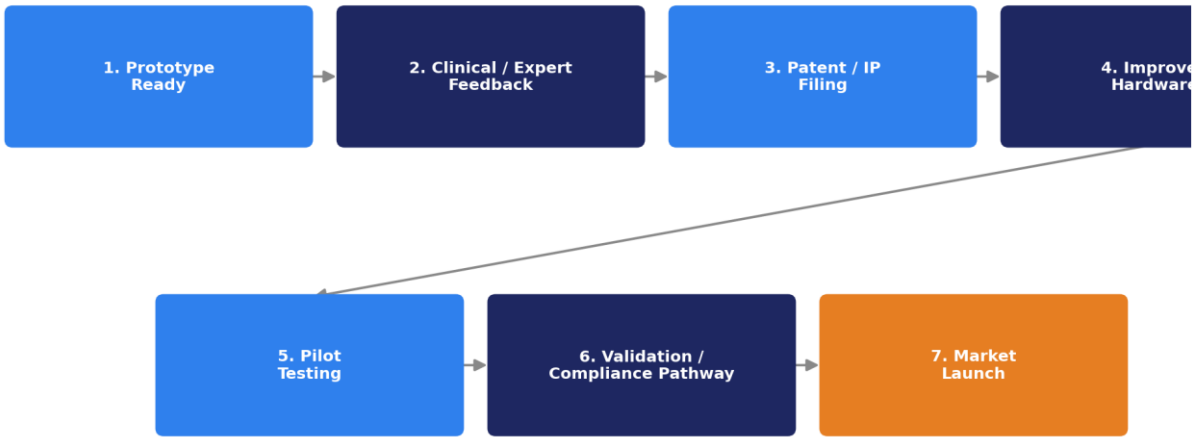


Fig. 6 — Seven-stage development roadmap

a) **Prototype Ready:** Working hardware and dashboard (current stage)

b) **Clinical/Expert Feedback:** Gathering input from counsellors, psychologists, and doctors

c) **Patent/IP Filing:** Protecting the innovation and software workflow

d) **Improved Hardware:** Refining enclosure, signal quality, portability, and report workflow

e) **Pilot Testing:** Deploying pilot units in clinics, wellness centers, and colleges

f) **Validation/Compliance Pathway:** Moving toward safety documentation and a possible ISO/CDSCO regulatory pathway with structured evaluation

g) **Market Launch:** Launching as a scalable, affordable product through institutional and direct channels

➢ Why This Roadmap Matters

It improves reliability, builds trust through testing and feedback, enables affordable deployment across India, and creates a credible pathway from student prototype to real-world impact.

➢ What the Team Is Asking For

- Clinical validation and expert mentorship
- Pilot testing partners
- Regulatory and certification guidance
- Incubation / innovation support
- Strategic funding and ecosystem connections

➢ Long-Term Vision

- **Accessible Care:** Making mental wellness support affordable and within reach
- **Data-Driven Counselling:** Empowering counsellors with objective, session-wise biofeedback insights
- **Smart Bio-Signal Analytics:** AI-assisted insights for stress, arousal, recovery, and progress tracking
- **Made in India / Scalable Globally:** Indigenous innovation with potential for local impact and global scale

10. Limitations and Assumptions

As with any early-stage prototype, MannPulseAI carries a number of limitations and working assumptions that should be considered alongside its stated benefits. Being transparent about these boundaries is important both for setting correct expectations with future pilot partners and for guiding the team's own development priorities.

➢ Current Limitations

- The system has not undergone formal clinical validation and should not be used to make diagnostic or treatment decisions in its current form.
- Signal quality can be affected by factors such as excessive movement, poor sensor contact, or extreme ambient temperature and humidity.
- The 16-feature extraction pipeline and TRR scoring model have been tuned on a limited testing dataset gathered from volunteer sessions, rather than a large, clinically representative sample.
- The current prototype requires a laptop to host the dashboard; a fully standalone or mobile version has not yet been developed.
- Long-term data storage, multi-user account management, and cloud synchronization are not yet implemented.

➢ Working Assumptions

- Users (counsellors) are assumed to have basic familiarity with operating a laptop-based application.
- Clients are assumed to consent to wearing a finger sensor for the duration of a counselling session.
- The system is assumed to be used as a complement to, not a substitute for, a trained counsellor's judgment.
- Hardware component costs are based on current market prices for commonly available parts and may vary with sourcing location and quantity.

These limitations directly inform the roadmap presented in the previous section — particularly the emphasis on clinical/expert feedback, pilot testing, and a formal validation and compliance pathway before any broader claims about diagnostic or clinical accuracy are made.

11. Conclusion

MannPulseAI, developed by Team Debuggers Squad, demonstrates a working, tested pathway from a clearly defined problem — the affordability and objectivity gap in India's counselling and mental health ecosystem — to a low-cost (₹2,700), portable, AI-powered GSR/EDA biofeedback prototype. The system combines a finger-sensor hardware unit, real-time signal processing, machine learning-based TRR scoring (Random Forest, Ridge Regression, EWMA), and automated PDF reporting into a workflow that is demo-ready, publicly validated (2.4M+ views on demonstration content), and grounded in established psychophysiology and biofeedback principles.

While explicitly positioned as a non-diagnostic, research/counselling-support tool rather than a certified medical device, the project lays out a credible, staged roadmap — from clinical feedback and IP protection through pilot testing and regulatory pathway exploration to eventual market launch — that could allow it to scale from a student prototype into an affordable, India-first, globally scalable biofeedback platform for counselling clinics, medical colleges, and research institutions.

12. References

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m) Databases consulted: World Health Organization (WHO), PubMed, PubMed Central (PMC), IEEE Xplore Digital Library.

13. Appendix

➢ Team: Debuggers Squad

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